

OPERATIONS RESEARCH AND SYSTEMS ANALYSIS

Linear programming – Tutorial 1

Task 1

- Toy Factory has to decide what wooden toys will be produced on one of their manufacturing line. The line can produce three types of the wooden toys: elephants, ponies and giraffes. The production cycle allows producing only a thousand of pieces and the line can processed only 240 kg of material in one production cycle. Each type of the wooden toy has a different material requirement, see the Table 1.
- Toy Factory is willing to maximize the profit with already signed contract for at least 200 giraffes. Each toy can be sold for following price: elephant for \$6, pony for \$7 and giraffe for \$4. What producing plan should be set?

Toy	Material (kg)
Elephant	0.2
Pony	0.3
Giraffe	0.5

- Simplex table and the solution is in the Moodle. TASK.xlsx
- **Do not forget to change the MS Excel language to English**

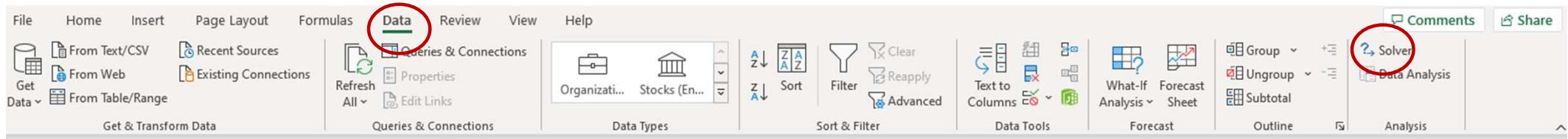
Task 1

□ Questions:

- a) What will be the change of the optimal solution if we produce 100 pcs of ponies?
- b) What will be the change of the total profit?
- c) What is the maximal number of ponies that can be produced without changing the optimal basis?
- d) How will change the solution if the supply of material goes down by 10 kg? and if goes up?
- e) Calculate the maximal change in the availability of material (without changing the basis)!
- f) What must be the minimal price of the ponies so that they worth producing?

Task 2

- Solve the problem from task 1 in solver



- If the Add-in **isn't** active, then:
 1. click **File**
 2. click **Options**
 3. click **Add-ins**
 4. choose Excel Add-Ins (on the bottom) and click **Go...**
 5. mark **Solver Add-in** and click **OK**

	A	B	C	D	E	F	G	H
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11		Prices	6	7	4			
12								
13			elephants	girafes	ponies	LHS	RHS	
14		wood	200	300	500	0	240	
15		giraffes	0	0	1	0	0.2	
16		total cap.	1	1	1	0	1	
17								
18					Profit		0	
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Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

SF\$14 <= \$G\$14

SF\$15 >= \$G\$15

SF\$16 <= \$G\$16

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method: **Simplex LP**

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Solver Results

Solver found a solution. All Constraints and optimality conditions are satisfied.

☒ Keep Solver Solution

☐ Restore Original Values

Reports

Answer

Sensitivity

Limits

☐ Return to Solver Parameters Dialog

☐ Outline Reports

Task 2



1. Interpret the solution from the solver output
2. Increase the amount of the wood that it will change the basis (what is the minimum?) and solve with the Solver again.
3. Change the amount of the wood back to 240. Change the price of the elephants to 4 and solve again.



Thank you for your attention.